

Hand Gesture Recognition

30 Days Complete Roadmap (Python + MediaPipe)

Tools & Requirements

Python 3.7+ | OpenCV | MediaPipe | NumPy | Webcam

Install: `pip install opencv-python mediapipe numpy`

Key Files: `hand_gesture.py` | `hand_landmarker.task` (MediaPipe model)

Key Code Components (from `hand_gesture.py`)

Function/Class	Purpose	Used In
<code>dist()</code> , <code>dist_2d()</code> , <code>angle_3pts()</code>	Geometry helpers	Week 1
<code>is_finger_up()</code> , <code>is_finger_curled()</code>	Finger state detection	Week 1-2
<code>is_finger_hooked()</code>	X sign detection	Week 2
<code>get_finger_states()</code>	All 5 fingers state	Week 2
<code>thumb_beside_fist()</code> , <code>thumb_is_across_palm()</code>	A, B, S signs	Week 2
<code>fingertips_touching_thumb()</code> , <code>thumb_index_circle()</code>	F, D signs	Week 2
<code>hand_pointing_sideways()</code> , <code>hand_pointing_down()</code>	G, H, P, Q signs	Week 2
<code>index_middle_crossed()</code>	R sign detection	Week 2
<code>classify_asl_letter()</code> - 11 Groups	Full A-Z classifier	Week 2
<code>LetterSmoother</code> (buffer=7 frames)	Flicker prevention	Week 3
<code>Settings</code> class	UI settings & low-light	Week 3
<code>apply_low_light_enhancement()</code>	Auto brightness + tint	Week 3
<code>draw_hand_landmarks()</code>	Skeleton overlay	Week 3
<code>draw_status_panel()</code>	Top-left info panel	Week 3
<code>draw_reference_panel()</code>	Right-side A-Z guide	Week 3
<code>draw_settings_panel()</code>	Center settings UI	Week 3
<code>draw_rounded_rect()</code>	Semi-transparent UI boxes	Week 3

ASL Fingerspelling Signs (A-Z) - `classify_asl_letter()` Groups

Group	Letters	Key Detection Logic
1	Y, I/J	Unique finger combos (pinky/thumb+pinky)
2	G, H	<code>hand_pointing_sideways()</code> + finger count
3	P, Q	<code>hand_pointing_down()</code> variants
4	L	Thumb + index up, L-shape
5	F	<code>thumb_index_circle()</code> + 3 fingers up
6	D, X, K, R, U, V	<code>index_up</code> combos, hooked, crossed, tip distance
7	W	Index + middle + ring up, spread
8	B	4-5 fingers up, thumb down
9	O	<code>fingertips_touching_thumb()</code> >= 2
10	C, E	Curved/curled, <code>angle_3pts()</code>
11	A, S, T, M, N	Fist variations, thumb position (z-depth, y-pos)

Week 1: Environment Setup & MediaPipe Basics

Day 1:

- Install Python 3.7+, OpenCV, MediaPipe, NumPy
- Test webcam: `cap = cv2.VideoCapture(0)`
- Study `hand_gesture.py` top-level structure & imports

Day 2:

- OpenCV basics: `cap.read()`, `cv2.imshow()`, `cv2.flip(frame, 1)`
- Understand BGR color format vs RGB (MediaPipe needs RGB)
- Draw circles & lines on video frames with `cv2.circle()`, `cv2.line()`

Day 3:

- Learn MediaPipe HandLandmarker: `HandLandmarkerOptions`, `RunningMode.IMAGE`
- Understand 21 hand landmarks (WRIST=0, fingertips=4,8,12,16,20)
- Detect hand landmarks from webcam & print landmark coords

Day 4:

- Study landmark IDs: `THUMB_TIP=4`, `INDEX_TIP=8`, `MIDDLE_TIP=12`, etc.
- Understand `TIP_IDS`, `DIP_IDS`, `PIP_IDS`, `MCP_IDS` arrays
- Practice identifying each finger point by drawing them on screen

Day 5:

- Learn `dist(a, b)`: 3D Euclidean distance between landmarks
- Learn `dist_2d(a, b)`: 2D distance (x,y only, ignores depth)
- Learn `angle_3pts(a, b, c)`: angle at b formed by ba and bc segments

Day 6:

- Study `is_finger_up()`: `tip.y < pip.y` (y increases downward in image)
- Study `is_finger_curled()`: `tip.y > pip.y` OR tip near MCP
- Study `is_finger_hooked()`: tip below DIP but DIP above PIP (for X sign)

Day 7:

- Review Week 1 - test landmark detection stability
- Practice: draw all 21 landmark dots with labels
- Fix any webcam / model file (`hand_landmarker.task`) issues

Week 2: Gesture Detection & Classification

Day 8:

- Study `get_finger_states()`: returns [thumb, index, middle, ring, pinky] booleans
- Understand handedness ('Left'/'Right') affects thumb detection (x-axis flipped)
- Practice: print finger states live on screen for all 5 fingers

Day 9:

- Study `thumb_beside_fist()`: thumb_y between `idx_mcp_y` and `idx_pip_y`
- Study `thumb_is_across_palm()`: thumb near palm center (for B sign)
- Practice detecting A (fist+thumb side), B (4 fingers up), S (thumb over fist)

Day 10:

- Study `fingertips_touching_thumb()`: counts tips within 0.06 threshold
- Study `thumb_index_circle()`: `dist(THUMB_TIP, INDEX_TIP) < 0.05`
- Practice detecting O (tips touch O shape), F (OK + 3 up), D (index + thumb circle)

Day 11:

- Study `hand_pointing_sideways()`: `abs(dx) > abs(dy) * 1.2` for index finger
- Study `hand_pointing_down()`: `MIDDLE_TIP.y > WRIST.y`
- Practice detecting G (sideways index), H (sideways index+middle), P, Q (pointing down)

Day 12:

- Study `index_middle_crossed()`: $\text{tip dist} < \text{mcp dist} * 0.5$ (for R)
- Understand `idx_mid_tip` distance: $< \text{palm} * 0.22 = \text{U}$ (together), $\geq = \text{V}$ (spread)
- Practice detecting R (crossed), U (together), V (peace), K (thumb between)

Day 13:

- Study `full_classify_asl_letter()` - all 11 classification groups in order
- Understand priority: Y/I first (unique), then sideways, downward, L, F, etc.
- Understand fist group: T (thumb between `idx+mid`), M (thumb under 3), N, S, A

Day 14:

- Test all 26 ASL letters A-Z with the classifier
- Note confused pairs: U vs V, M vs N, A vs S, G vs H, P vs Q
- Document accuracy issues and threshold adjustments needed

Week 3: UI & Visualization

Day 15:

- Study `draw_hand_landmarks()`: converts normalized coords to pixel coords
- Practice drawing skeleton with white lines + green dots on frame
- Customize landmark colors (green UP, red DOWN)

Day 16:

- Study `draw_status_panel()`: responsive scale = $h/480.0$ pattern
- Understand `draw_rounded_rect()` with alpha blending (`cv2.addWeighted`)
- Practice: modify panel layout, add custom text, change colors

Day 17:

- Study `draw_reference_panel()`: `SIGN_INSTRUCTIONS` dict, current letter highlighting
- Understand `ASL_DESCRIPTIONS` dict for detailed sign descriptions
- Add custom descriptions or emoji-style indicators

Day 18:

- Study `LetterSmoother` class: buffer of 7 frames, Counter for most common
- Requires $\geq 40\%$ agreement (2+ out of 7) to change displayed letter
- Test: change `buffer_size` (3=faster, 12=smoother), observe responsiveness

Day 19:

- Study `main()` function: `HandLandmarkerOptions`, `RunningMode.IMAGE`
- Understand `Settings` class: bg presets, `auto_low_light`, `manual_gain`
- Study `apply_low_light_enhancement()`: `auto_gain` ramp + BGR tint mask

Day 20:

- Study `BG_PRESETS`: 7 color options (Warm Amber, Soft Cyan, Mint, etc.)
- Add keyboard controls: r=reference, s=settings, b=boost, [/]=gain, 1-7=colors
- Add new UI feature: collected letter string display, frame counter

Day 21:

- Test full UI on different screen sizes (480p, 720p, 1080p)
- Verify responsive scale = $h/480.0$ works for all panel sizes
- Review all visual elements, fix any overlapping UI panels

Week 4: Improvement & Final Project

Day 22:

- Improve M, N, T classification: tune `thumb_y` vs `pip_y` thresholds
- Fine-tune `close` = `palm_size` * 0.3 for touching detection
- Test edge cases: different hand sizes, distances, angles

Day 23:

- Add word building: collect detected letters into words string

- Add backspace gesture (e.g., hold 'D' sign for 30 frames)
- Display built word on screen with space gesture support

Day 24:

- Add quiz/practice mode: show random target letter on screen
- Compare smoother.current with target letter each frame
- Give visual feedback: green flash = correct, red = incorrect

Day 25:

- Add score tracking: correct_count, total_attempts, accuracy %
- Display live accuracy percentage and streak counter
- Save scores to JSON file for session history

Day 26:

- Performance optimization: profile with cProfile, reduce copies
- Test on different lighting (dark room, bright sunlight)
- Tune auto_gain ramp: brightness < 90 triggers boost up to 2.0x

Day 27:

- Error handling: webcam not found (cap.isOpened() check)
- Handle missing hand_landmarker.task with download URL message
- Add try/except around landmark detection for robustness

Day 28:

- Record demo video: test all 26 letters A-Z on camera
- Demonstrate settings panel (s key), reference guide (r key)
- Show low-light boost (b key) and background color presets (1-7)

Day 29:

- Update README.md: installation, usage, controls, screenshots
- Add annotated screenshots of UI panels and landmark overlay
- Write project report: architecture, challenges, accuracy results

Day 30:

- Final review: test all features, edge cases, error handling
- Push final version to GitHub with clean commit history
- Submit project! Share demo with community

Keyboard Controls Reference

Key	Action
q	Quit the application
r	Toggle ASL reference guide panel (right side)
s	Toggle settings panel (center overlay)
b	Toggle auto low-light brightness boost
[/]	Decrease / Increase manual brightness gain (0.5x - 2.5x)
- / +	Decrease / Increase background tint strength (0% - 90%)
1 . . 7	Select background color preset (Off / Amber / Cyan / Mint / Lavender / Rose / Gray)
n	Cycle to next background color